

Week 1

August 27/29

- Course introduction
 - Cover Syllabus / University Policies
- History/What is Computer Science
 - Problem Solving
 - Discussed a well-structured general problem format (INPUT, PROBLEM SOLVING STEPS, OUTPUT)
 - emphasis on important people (Babbage, Lovelace, Turing, etc..)
 - generations of computers (vacuum tubes → quantum/molecular computers)

Week 2

September 3/5

- Modern computer components
 - CPU, Memory, Storage, I/O devices, etc...
 - Brief introduction on how these components work together
- Number Systems
 - Decimal, Binary, Hexadecimal
 - Conversions between these bases, and how they relate to computers

Week 3

September 10/12

- Binary Arithmetic
 - Binary Addition
- Binary Arithmetic
 - Two's Complement and Binary Subtraction

Week 4

September 17/19

- Boolean Algebra
 - Boolean Algebra Rules and Laws
 - Boolean Expressions
- Gates and Circuits
 - Boolean Logic Gates
 - Boolean Circuits

Week 5

September 24/26

- **Test 1**
 - CS History, Computer Components, Number Systems, Binary Arithmetic, Boolean Algebra, Logic Circuits
- Data Representation
 - Combining the ideas of number systems, binary arithmetic, and boolean logic under the umbrella of computers.

Week 6

October 1/3

- Algorithms
 - Problem Solving
 - Program Flow (Flowcharts)
- Introduction to Programming
 - Using Repl.it online programming environment
 - "Hello World"
 - Simple input/output.

Week 7

October 8/10

- Control Structure: If/Elif/Else
 - If/Elif/Else Statements
 - Nested If/Elif/Else Statements
 - Multi-Conditional If/Elif/Else Statements
- If/Elif/Else Lab/Assignment
 - Movie Recommendation System

Week 8

October 15/17

- Control Structure: Loops
 - For Loops
 - While Loops
- Loops Lab/Assignment
 - Printing out every even, 3rd, and 5th number between 1 and 100.

Week 9

October 22/24

- Functions
 - Defining functions
 - Calling functions
 - Passing arguments
 - Returning values
- Functions Lab/Assignment
 - Define a function that takes two integers as arguments (base, exponent), and uses a FOR loop and multiplication to compute $\text{base}^{\text{exponent}}$.

Week 10

October 29/31

- Functions (cont.)
 - main() function
 - scope
- Functions (cont.) Lab/Assignment
 - Menu program